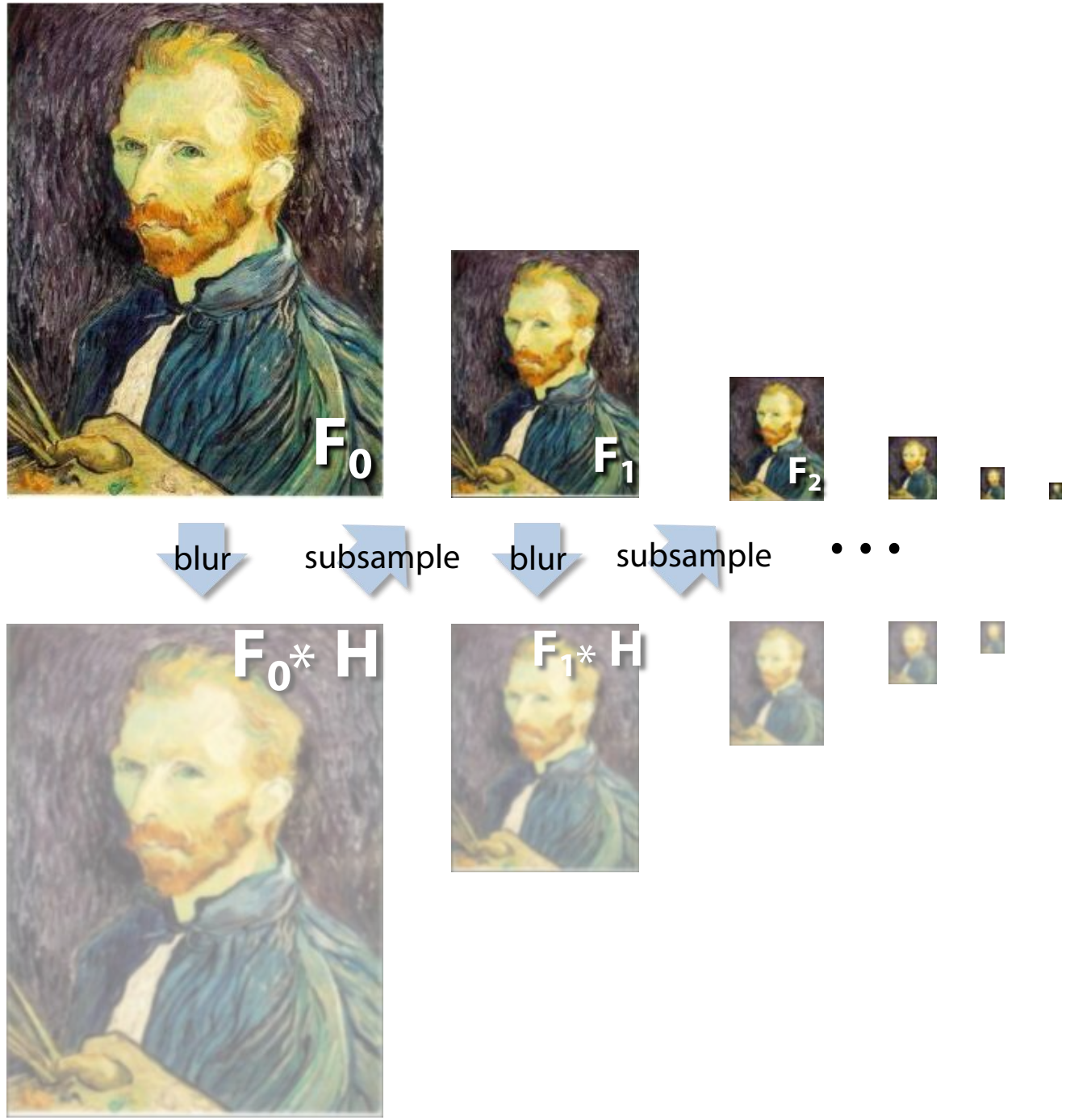
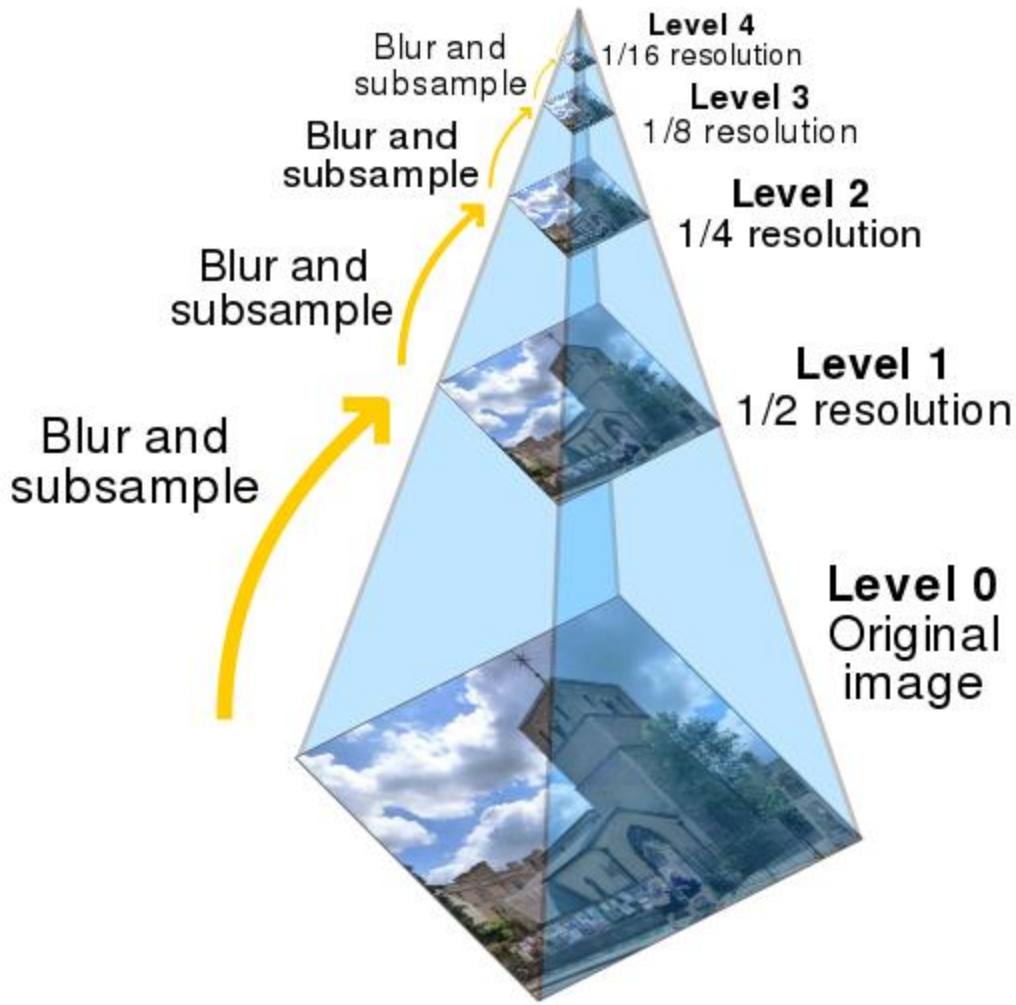
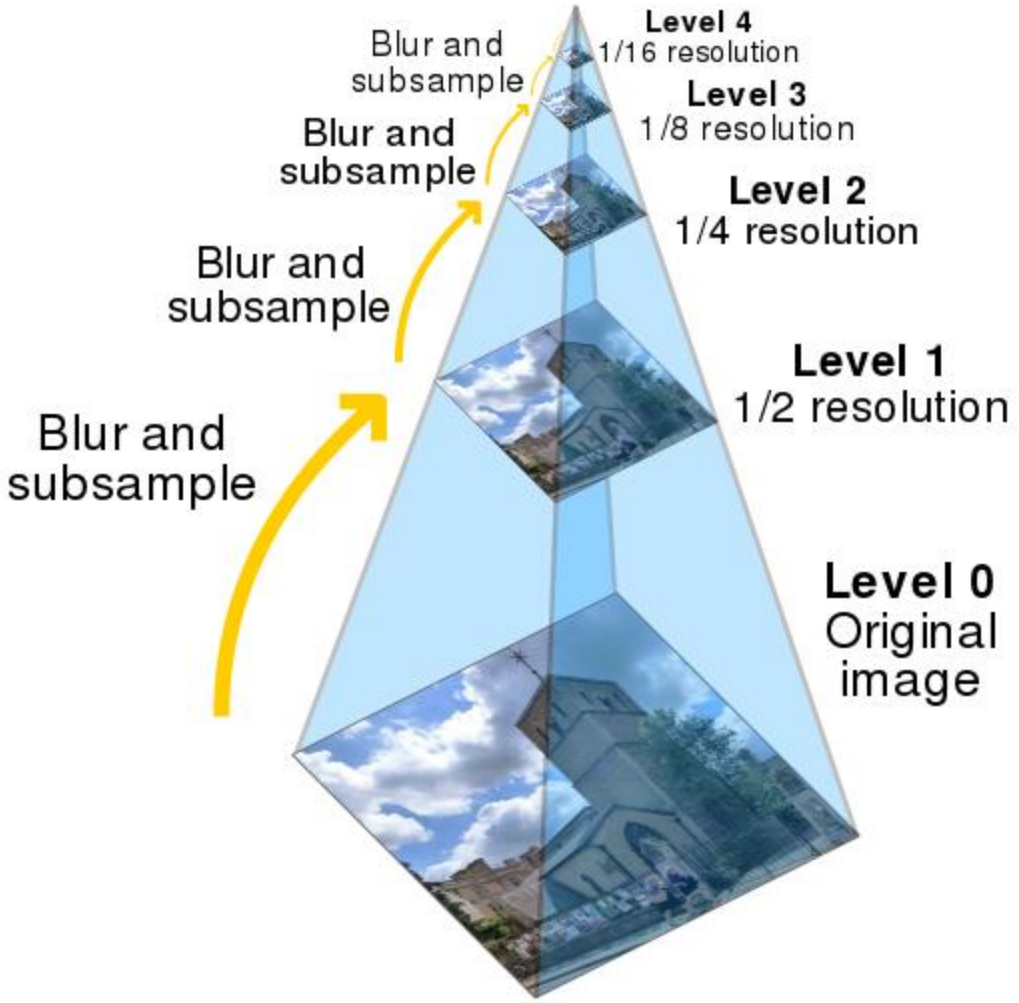
Gaussian pyramid




Blur and subsample
Blur and subsample
Blur and subsample
Blur and subsample
Level 4
1/16 resolution
Level 3
1/8 resolution
Level 2
1/4 resolution
Level 1
1/2 resolution
Level 0
Original image

Gaussian pyramids [Burt and Adelson, 1983]

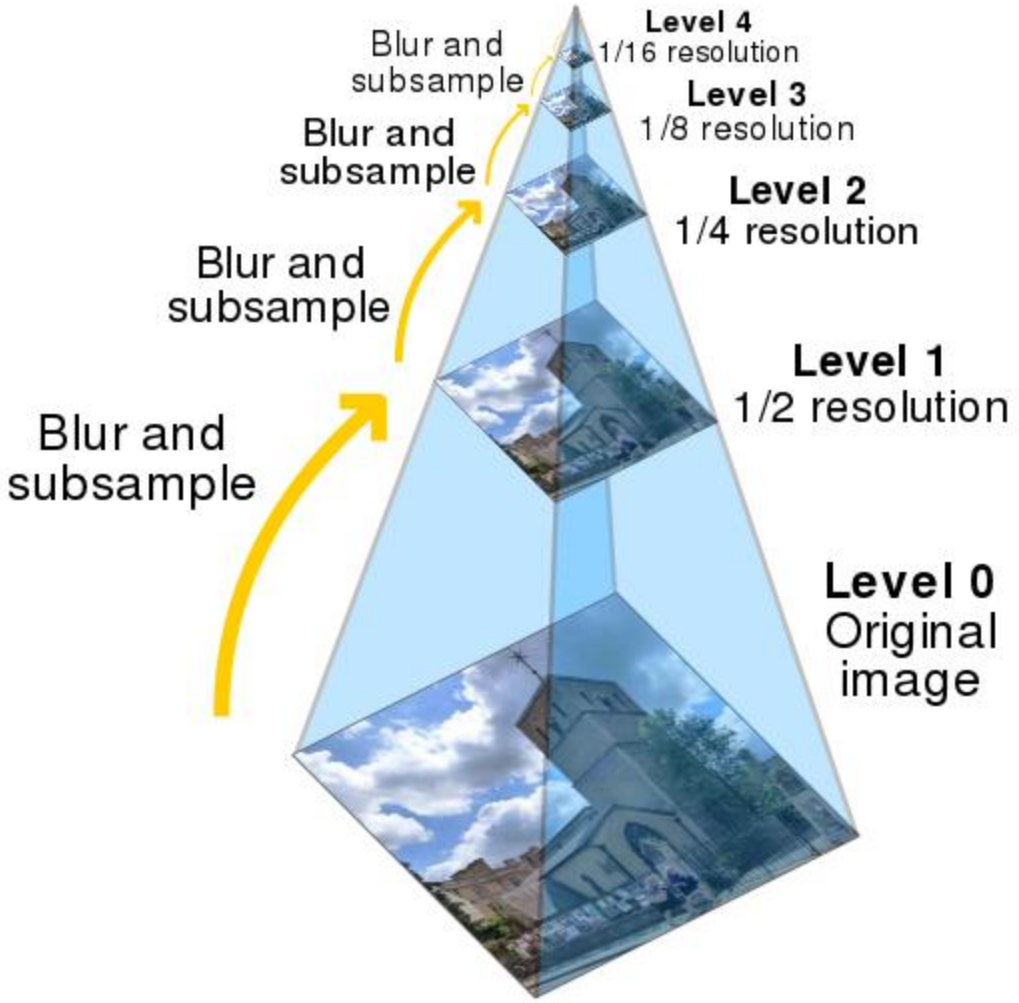


- In computer graphics, a *mip map* [Williams, 1983]
- A precursor to *wavelet transform*

Gaussian Pyramids have all sorts of applications in computer vision

Source: S. Seitz

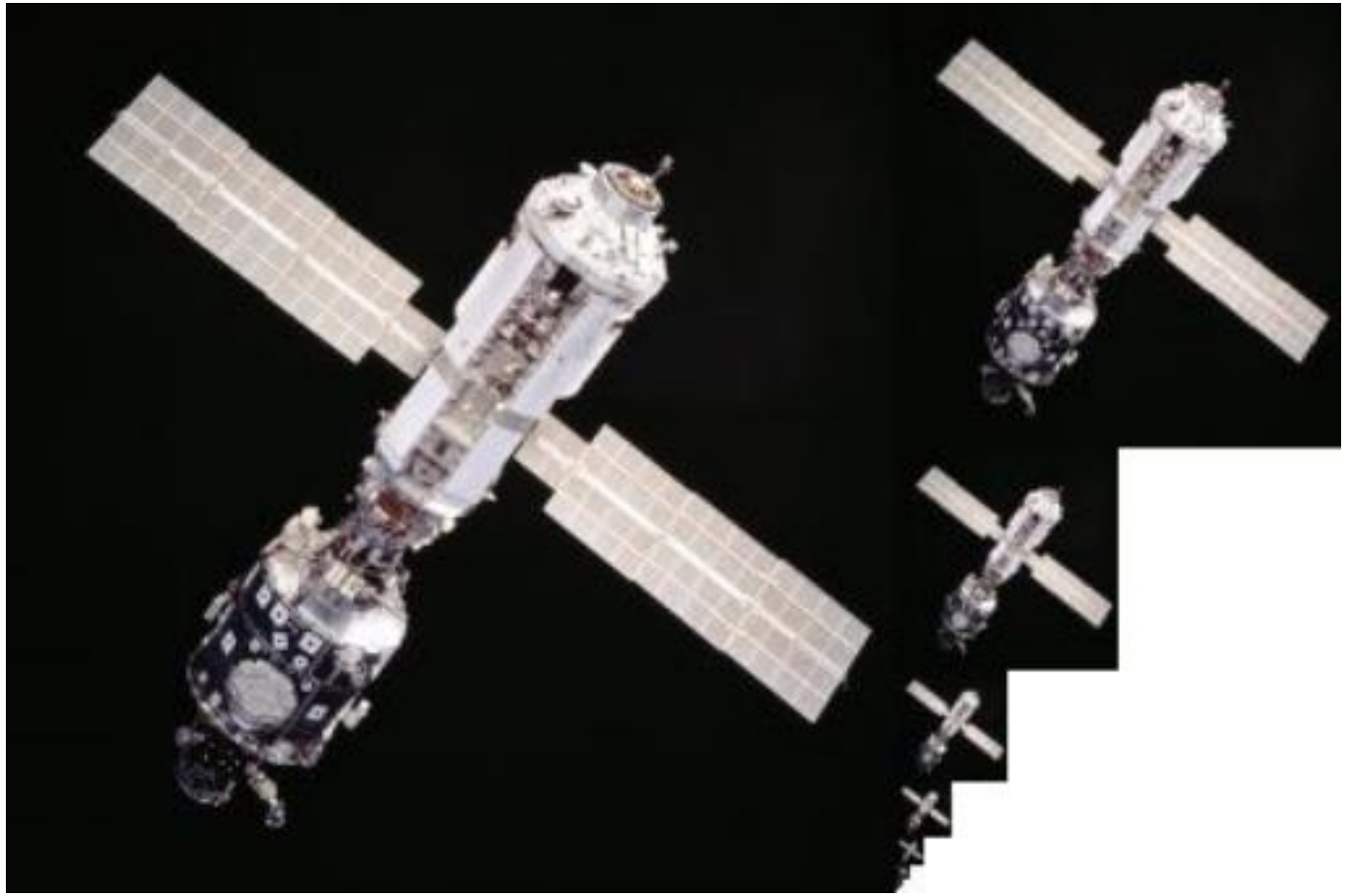
Gaussian pyramids [Burt and Adelson, 1983]



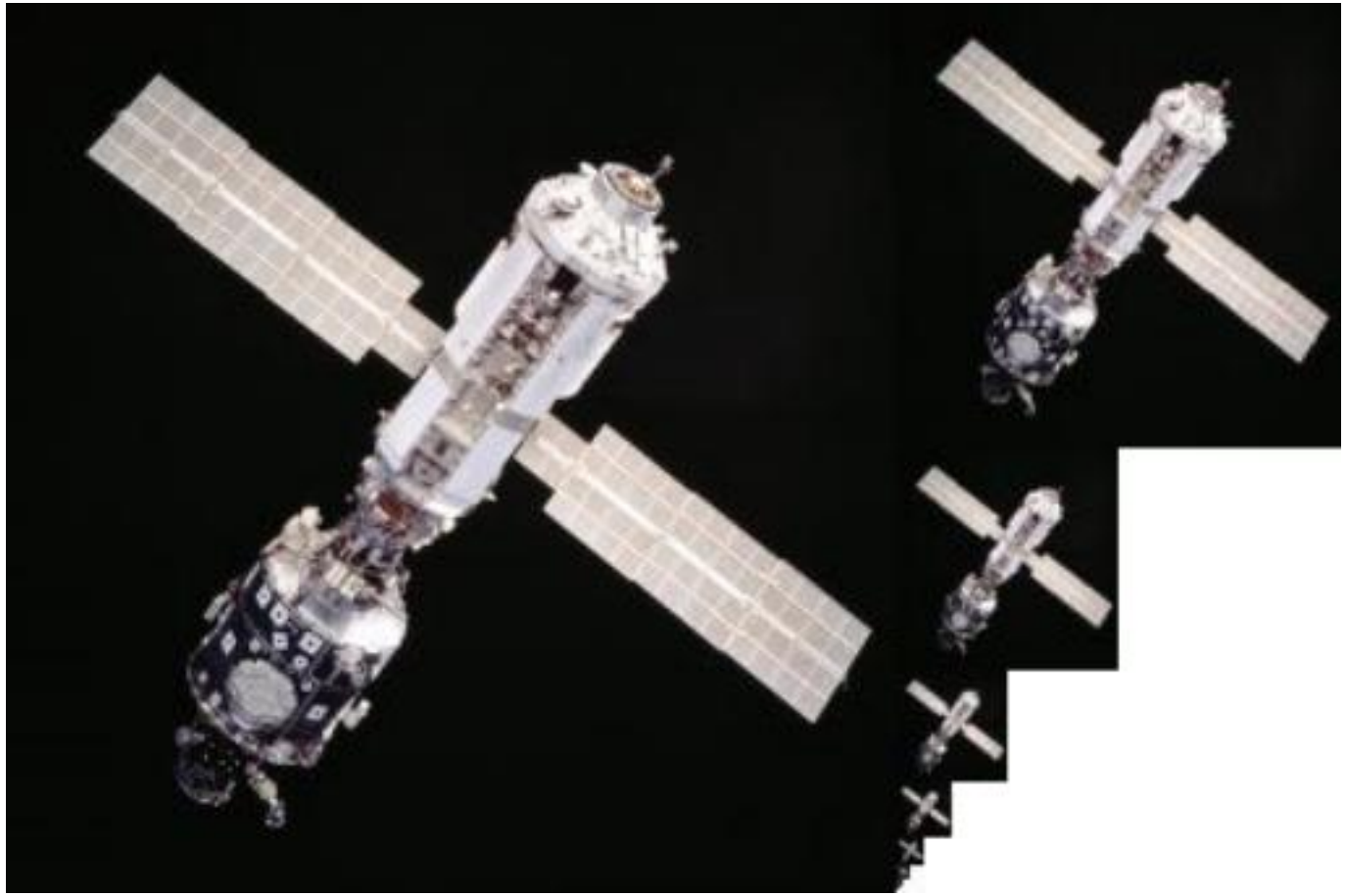
Q: How much space does a Gaussian pyramid take compared to the original image?

Source: S. Seitz

Gaussian pyramid



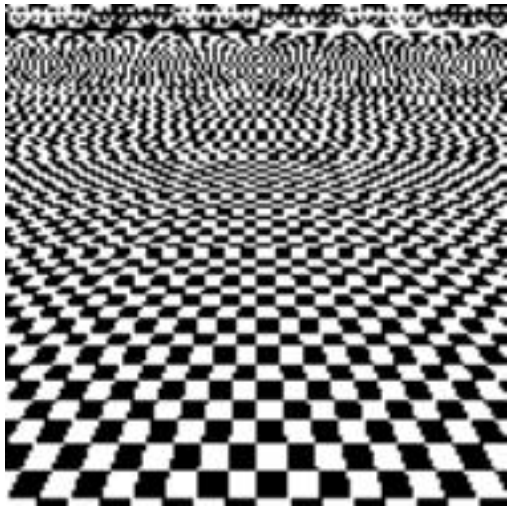
Gaussian pyramid



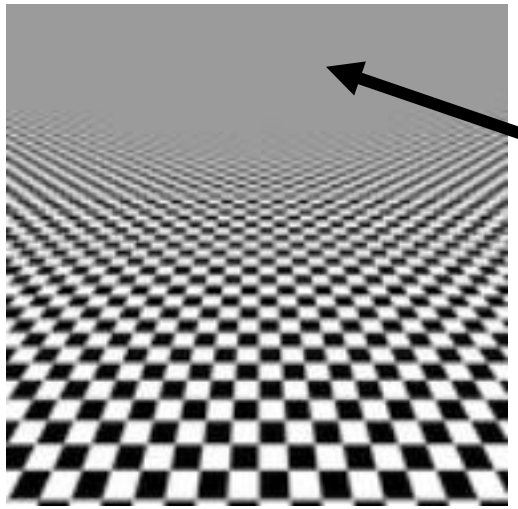
Answer: 4/3 the amount of space as the original image alone
See https://en.wikipedia.org/wiki/Geometric_series

Back to the checkerboard

- What should happen when you make the checkerboard smaller and smaller?



Naïve subsampling



Proper prefiltering
("antialiasing")

Image turns grey!
(Average of black and white squares, because each pixel contains both.)

Questions?